

What are hyperfunctors isomorphic to?

OPG-59973

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Let $\mathfrak{A}$ be an indexed family of sets.

Products are $\prod A$ for $A \in \prod \mathfrak{A}$.

Hyperfunctors are filters $\mathfrak{F}\Gamma$ on the lattice Γ of all finite unions of products.

Problem Is $\bigcap^{\text{FCD}}$ a bijection from hyperfunctors $\mathfrak{F}\Gamma$ to:

- prestaroids on $\mathfrak{A}$;
- staroids on $\mathfrak{A}$;
- completary staroids on $\mathfrak{A}$?

If yes, is up^Γ defining the inverse bijection? If not, characterize the image of the function $\bigcap^{\text{FCD}}$ defined on $\mathfrak{F}\Gamma$.

Consider also the variant of this problem with the set Γ replaced with the set Γ^* of complements of elements of the set Γ .

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>